

R Programming Tutorial (Part 2)

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Outline

- Interval Estimate in R
- Hypothesis Testing in R

Interval Estimate in R for Proportions

Example

- A recent study investigated the effects of a "Buckle Up Your Toddlers" campaign to get parents to use the grocery cart seat belts. Investigators observed a representative sample of parents at grocery stores in a large city, and found 192 out of 594 parents buckling up their toddlers.
- a. Compute a point estimate of the true proportion of all parents who buckle up their toddlers.
- b. Construct and interpret a 95% confidence interval for, the true proportion of all parents who buckle up their toddlers.

$\hat{p}$
↑
pba

t) / n)

)8448

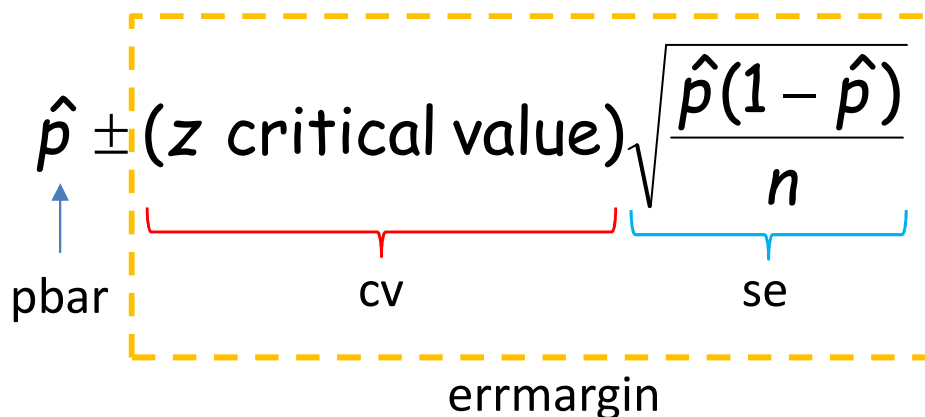
Interval Estimate in R for Proportions

WHAT WE KNOW?

- Sample size, n: 594
- 192 of the sample buckling up their toddlers.

WHAT WE WANT TO FIND?

- Point estimate of the true proportion.
- 95% confidence interval for the true proportion.



$$\hat{p} \pm (z \text{ critical value}) \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Diagram labels: $\hat{p}$ (point estimate), cv (critical value), se (standard error), $errmargin$ (error margin).

```

1      n = 594
2      k = 192
3      pbar = n/k;
4      se = sqrt(pbar*(1-pbar)/n)
5      cv = qnorm(0.975)
6      errmargin = se*cv;
8
9      // Get the interval
10     pbar+c(-errmargin,
errmargin)

```

```
[1] 0.2856198 0.3608448
```

Interval Estimate in R for Proportions

$1 - \alpha$	α	$\alpha / 2$	$z_{\alpha/2}$
.90	.10	.05	$z_{.05} = 1.645$
.95	.05	.025	$z_{.025} = 1.96$
.98	.02	.01	$z_{.01} = 2.33$
.99	.01	.005	$z_{.005} = 2.575$

```

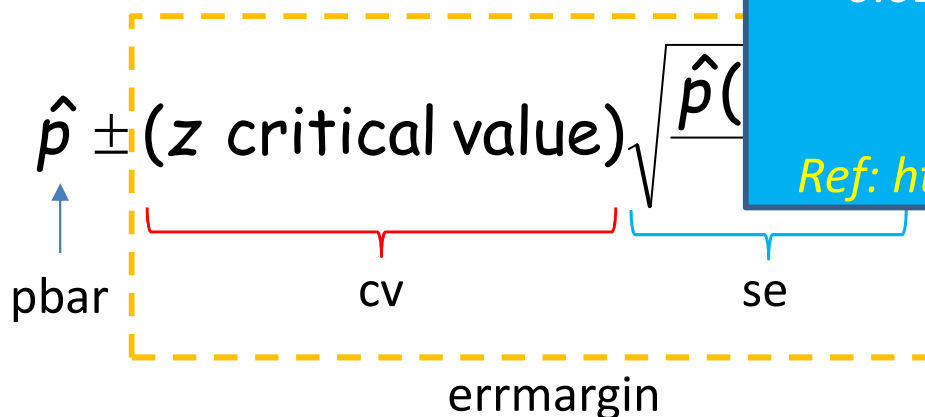
1      n = 594
2      k = 192
3      pbar = n/k;
4      se = sqrt(pbar*(1-pbar)/n)
5      cv = qnorm(0.975)
6      errmargin = se*cv;
7
8
9      // Get the interval
10     pbar - errmargin

```

- 95% confidence interval for the true proportion.

qnorm() can be used to get the Z critical value. E.g.: for 95% confidence level, $\alpha/2 = 0.025$, however, parameter to put in ***qnorm()*** is $1 - 0.025 = 0.975$

Ref: <http://seankross.com/notes/dpqr/>



```

17
18
19
20
21
22

```

```
[1] 0.2856198 0.3608448
```

Interval Estimate in R for Proportions

WHAT WE KNOW?

- Sample size, n : 594
- 192 of the sample buckled up their toddlers.

WHAT WE WANT TO FIND

- Point estimate of the proportion.
- 95% confidence interval for the true proportion.

$$\hat{p} \pm (z \text{ critical value})$$

↑
pbar

cv

errm

1 $n = 594$

www.utm.my

Solution

a) point estimate: $p = 192 / 594 = 0.3232$

b) 95% CI: = 95% CI: $0.3232 \pm 1.960 \sqrt{\frac{0.3232(0.6768)}{594}} = (0.2856, 0.3608)$.

I am 95% confident that the true proportion of parents who buckle up their toddlers is between 0.286 and 0.361.

Interval Estimate in R for Proportions (Alternative)

`prop.test()` is used to test proportions.
 It includes the calculation of confidence interval (95% by default)

How to use `prop.test()`:

```
prop.test(x, n, p = NULL,
          alternative = c("two.sided", "less", "greater"),
          conf.level = 0.95, correct = TRUE)
```

Ref: <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/prop.test.html>

```
1      n = 594
2      k = 192
3
4      prop.test(k, n)
5
6
7
8
9
10
11
```

1-sample proportions test with continuity correction

```
data:  k out of n, null probability 0.5
X-squared = 73.537, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.2860486 0.3627384
sample estimates:
```

```
      p
0.3232323
```

Confidence Interval (Default: 95%)

Interval Estimate in R for Mean

WHAT WE KNOW?

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Example

- A computer company samples demand during lead time over 25 time periods:

235	374	309	499	253
421	361	514	462	369
394	439	348	344	330
261	374	302	466	535
386	316	296	332	334

It is known that the standard deviation of demand over lead time is 75 computers. We want to estimate the **mean** demand over lead time with 95% confidence in order to set inventory levels

1 n = 25

```
75
, 374, 309, 499,
, 361, 514, 462,
, 439, 348, 344,
, 374, 302, 466,
, 316, 296, 332,
```

```
an(x)
ev/sqrt(n)
m(0.975)
= cv*se
```

```
I.
rrmargin,
```

```
7605 399.5595
```


Interval Estimate in R for Mean

WHAT WE KNOW?

- Sample size, n: 25
- Standard deviation: 75

WHAT WE WANT TO FIND?

- 95% confidence interval for the mean.

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Diagram illustrating the components of the confidence interval formula:

- $\bar{x}$ is labeled as **xbar**.
- $z_{\alpha/2}$ is labeled as **cv** (critical value).
- $\frac{\sigma}{\sqrt{n}}$ is labeled as **se** (standard error).
- The entire term $z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ is labeled as **errmargin** (error margin).
- σ is labeled as **stdDev** (standard deviation).

```

1      n = 25
2      stdDev = 75
3      x = c(235, 374, 309, 499,
4            253,
5            421, 361, 514, 462,
6            369,
7            394, 439, 348, 344,
8            330,
9            261, 374, 302, 466,
10           535,
11           386, 316, 296, 332,
12           334)
13
14     xbar = mean(x)
15     se = stdDev/sqrt(n)
16     cv = qnorm(0.975)
17     errmargin = cv*se
18
19     // 95% C.I.
20     xbar+c(-errmargin,
21           errmargin)
22
23 [1] 340.7605 399.5595
  
```

Interval Estimate in R for Mean

WHAT WE KNOW?

- Sample size, n : 25
- Standard deviation

WHAT WE WANT TO FIND

- 95% confidence interval for mean.

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$\bar{x}$ is labeled as **xbar** with an arrow pointing to it.
 $z_{\alpha/2}$ is labeled as **cv** (critical value) with a bracket underneath.
 $\frac{\sigma}{\sqrt{n}}$ is labeled as **errmargin** (error margin) with a bracket underneath.

Solution

- In order to use our confidence interval estimator, we need the following pieces of data:

$\bar{x}$	370.16
$z_{\alpha/2}$	1.96
σ	75
n	25

Calculated from the data...

$1 - \alpha = .95, \therefore \alpha/2 = .025$

so $z_{\alpha/2} = z_{.025} = 1.96$

Given

therefore: $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 370.16 \pm z_{.025} \frac{75}{\sqrt{25}} = 370.16 \pm 1.96 \frac{75}{\sqrt{25}} = 370.16 \pm 29.40$

The **lower** and **upper** confidence limits are **340.76** and **399.56**.

Hypothesis Testing

Decision Criterion

■ Traditional method:

- If the test statistic falls within the critical region, **reject H_0** .
- If the test statistic does not fall within the critical region, **fail to reject H_0** .

Hypothesis Testing

METHOD 1

1. Identify critical region
2. Calculate test statistics
3. If test statistics fall within critical region, reject H_0
4. If test statistics does not fall within critical region, fail to reject H_0

METHOD 2

1. Define significance level.
2. Calculate test statistics.
3. Find p-value
4. If p-value < significance level, reject H_0
5. If p-value > significance level, fail to reject H_0

Hypothesis Testing in R

Example

- **Writing a Hit Song** In the manual “How to Have a Number One the Easy Way,” by KLF Publications, it is stated that a song “must be no longer than three minutes and thirty seconds” (or 210 seconds). A simple random sample of 40 current hit songs results in a mean length of 252.5 sec. (The songs are by Timberlake, Furtado, Daughtry, Stefani, Fergie, Akon, Ludacris, etc.) Assume that the standard deviation of song lengths is 54.5 sec. Use a 0.05 significance level to test the claim that the sample is from a population of songs with a mean greater than 210 sec. What do these results suggest about the advice given in the manual?

```
z test
) / (sd/sqrt(n))

critical value
orm(1-alpha)
```

Hypothesis Testing in R

Population mean with Known Variance – One Tailed (Traditional Method)

WHAT WE KNOW?

- Sample size, n : 40
- Standard deviation: 54.5
- Sample mean: 252.5
- $H_0: \mu = 210$ $H_1: \mu > 210$
- Significance level: 0.05

$$z = \frac{\bar{x} - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

```
1      n = 40
2      sd = 54.5
3      xbar = 252.5
4      mu = 210
5      alpha = 0.05
6
7      // Calculate z test
8statistics
9      z = (xbar-mu) / (sd/sqrt(n))
10
11     // Calculate critical value
12     z.alpha = qnorm(1-alpha)
13
14
15
16     > z
17     [1] 4.931993
18
19     > z.alpha
20     [1] 1.644854
21
22
```


Hypothesis Testing in R

Population mean with Known Variance – One Tailed (P-value Method)

WHAT WE KNOW?

- Sample size, n: 40
- Standard deviation: 54.5
- Sample mean: 252.5
- $H_0: \mu = 210$ $H_1: \mu > 210$
- Significance level: 0.05

$$Z = \frac{\bar{X} - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

```

1      n = 40
2      sd = 54.5
3      xbar = 252.5
4      mu = 210
5      alpha = 0.05
6
7      // Calculate z test
8      statistics
9      z = (xbar-mu) / (sd/sqrt(n))
10
11     // Calculate p-value of z
12     pval = pnorm(z,
13                 1.959963984540054,

```

pnorm() can be used to get p-value for a calculated test statistics.

Ref: <http://seankross.com/notes/dpqr/>

```

19
20     [1] 4.069748e-07
21
22

```


Hypothesis Testing in R

Population
Variance – σ^2
(P-value Method)

WHAT WE KNOW

- Sample size $n = 40$
- Standard deviation $\sigma = 54.5$
- Sample mean $\bar{x} = 252.5$
- $H_0: \mu = 210$
- Significant level $\alpha = 0.05$

Z =

original claim: $\mu > 210$ sec

$H_0: \mu = 210$ sec

$H_1: \mu > 210$ sec

$\alpha = 0.05$

C.V. $z = z_\alpha = z_{0.05} = 1.645$

calculations:

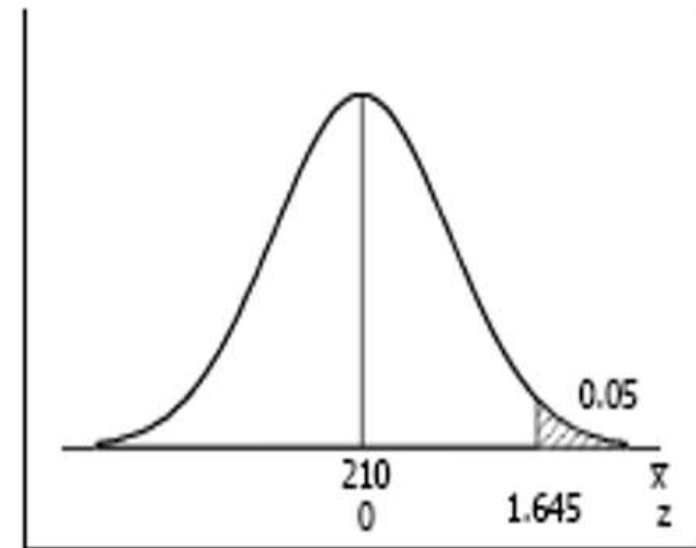
$$\begin{aligned} z_{\bar{x}} &= (\bar{x} - \mu_{\bar{x}}) / \sigma_{\bar{x}} \\ &= (252.5 - 210) / (54.5 / \sqrt{40}) \\ &= 42.5 / 8.6172 = 4.93 \end{aligned}$$

$$\text{P-value} = P(z > 4.93) = 0.5 - 0.4999 = 0.0001$$

conclusion:

Reject H_0 ; there is sufficient evidence to conclude that $\mu > 210$. There is sufficient evidence to support the claim that the sample is from a population of songs with a mean greater than 210 seconds. These results suggest that the advice given in the manual is not good advice.

Solution



Hypothesis Testing in R

Population mean with Known Variance – Two Tailed (Traditional Method)

Problem:

Suppose the mean weight of King Penguins found in an Antarctic colony last year was 15.4 kg. In a sample of 35 penguins same time this year in the same colony, the mean penguin weight is 14.6 kg. Assume the population standard deviation is 2.5 kg. At .05 significance level, can we reject the null hypothesis that the mean penguin weight does not differ from last year?

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Hypothesis Testing in R

Population mean with Known Variance – Two Tailed (Traditional Method)

WHAT WE KNOW?

- Sample size, n: 35
- Standard deviation: 2.5
- Sample mean: 14.6
- $H_0: \mu = 15.4$ $H_1: \mu \neq 15.4$
- Significance level: 0.05

$$Z = \frac{\bar{X} - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

```

1      n = 35
2      sd = 2.5
3      xbar = 14.6
4      mu = 15.4
5      alpha = 0.05
6
7      // Calculate Z statistics
8      z = (xbar-mu)/(sd/sqrt(n))
9
10     // Calculate critical value
11     z.alpha = qnorm(1-(alpha/2))
12
13     // Display the c.v. on both
14     sides
15     c(-z.alpha, z.alpha)
16
17     > Z
18     [1] -1.893146
19
20     > c(-z.alpha, z.alpha)
21     [1] -1.959964  1.959964
22
  
```

Hypothesis Testing in R

Population mean with Known Variance – Two Tailed (Traditional Method)

WHAT WE KNOW?

- Sample size, n : 35
- Standard deviation: 2.5
- Sample mean: 14.6
- $H_0: \mu = 15.4$ $H_1: \mu \neq 15.4$
- Significance level: 0.05

In two tailed test, the critical value is calculated based on half of the alpha value

$$\frac{\sigma}{\sqrt{n}}$$

```

1      n = 35
2      sd = 2.5
3      xbar = 14.6
4      mu = 15.4
5      alpha = 0.05
6
7      // Calculate Z statistics
8      z = (xbar-mu)/(sd/sqrt(n))
9
10     // Calculate critical value
11     z.alpha = qnorm(1-(alpha/2))
12
13     // Display the c.v. on both
14     sides
15     c(-z.alpha, z.alpha)
  
```

```

> z
[1] -1.893146
> c(-z.alpha, z.alpha)
[1] -1.959964  1.959964
  
```

Hypothesis Testing in R

Population mean with Known Variance – Two Tailed (P-value Method)

WHAT WE KNOW?

- Sample size, n : 35
- Standard deviation: 2.5
- Sample mean: 14.6
- $H_0: \mu = 15.4$ $H_1: \mu \neq 15.4$
- Significance level: 0.05

$$Z = \frac{\bar{X} - \mu_x}{\frac{\sigma}{\sqrt{n}}}$$

```
1      n = 35
2      sd = 2.5
3      xbar = 14.6
4      mu = 15.4
5      alpha = 0.05
6
7      // Calculate Z statistics
8      z = (xbar-mu)/(sd/sqrt(n))
9
10     // Calculate p-value for Z
11     pval = 2*pnorm(z)
12
13
14
15
16
17
18
19
20     [1] 0.05833852
21
22
```

Hypothesis Testing in R

Population mean with Known Variance – Two Tailed (P-value Method)

WHAT WE KNOW?

- Sample size, n : 35
- Standard deviation: 2.5
- Sample mean: 14.6
- $H_0: \mu = 15.4$ $H_1: \mu \neq 15.4$
- Significance level: 0.05

P-value calculated by `pnorm()` represent one-tailed P-value. For two tailed test, the P-value must multiply by 2

$$\frac{\sigma}{\sqrt{n}}$$

```

1      n = 35
2      sd = 2.5
3      xbar = 14.6
4      mu = 15.4
5      alpha = 0.05
6
7      // Calculate Z statistics
8      z = (xbar-mu)/(sd/sqrt(n))
9
10     // Calculate p-value for Z
11     pval = 2*pnorm(z)
12
13
14

```

```
[1] 0.05833852
```


Hypothesis Testing in R

Population mean with Unknown Variance – One Tailed (Traditional Method)

WHAT WE KNOW?

- Sample size, n: 15
- List of data points given
- $H_0: \mu = 0.82$ $H_1: \mu > 0.82$
- Significance level: 0.05

qt() can be used to get the T critical value.

2 parameters: 1) t statistics, 2) df

Ref: <http://stat.ethz.ch/R-manual/R-devel/library/stats/html/TDist.html>

```

1      x = c(0.8411, 0.8191,
2.8182,          0.8125,
3.8750, 0.8580,
4.8532, 0.8483, 0.8276,
5          0.7983, 0.8042,
6.8730,
7          0.8282, 0.8359,
8.8660)
9      n = 15
10     s = sd(x)
11     xbar = mean(x)
12     mu = 0.82
13
14     // Calculate t statistics
15     t = (xbar-mu) / (s/sqrt(n))
16
17     // Calculate critical value
18     alpha = 0.05
19
20     > t
21     [1] 2.718979
22     > t.alpha
23     [1] 1.76131
  
```

Hypothesis Testing in R

Example

The increased availability of light materials with high strength has revolutionized the design and manufacture of golf clubs, particularly drivers. Clubs with hollow heads and very thin faces can result in much longer tee shots, especially for players of modest skills. This is due partly to the “spring-like effect” that the thin face imparts to the ball. Firing a golf ball at the head of the club and measuring the ratio of the outgoing velocity of the ball to the incoming velocity can quantify this spring-like effect. The ratio of velocities is called the coefficient of restitution of the club. An experiment was performed in which **15 drivers** produced by a particular club maker were selected at random and their coefficients of restitution measured. In the experiment the golf balls were fired from an air cannon so that the incoming velocity and spin rate of the ball could be precisely controlled. **It is of interest to determine if there is evidence (with $\alpha = 0.05$) to support a claim that the mean coefficient of restitution exceeds 0.82.** The observations follow:

0.8411	0.8191	0.8182	0.8125	0.8750
0.8580	0.8532	0.8483	0.8276	0.7983
0.8042	0.8730	0.8282	0.8359	0.8660

```

1 x = c(0.8411, 0.8191,
      0.8125,
      0.8276,
      0.7983, 0.8042,
      0.8282, 0.8359,
      0.8580, 0.8532, 0.8483, 0.8660)

mean(x)
# [1] 0.8359

# calculate t statistics
t = (mean(x) - mu) / (s / sqrt(n))

# calculate critical value
# alpha = 0.05
# df = n - 1 = 14
# qt(0.95, 14)
# [1] 1.76131
  
```

Hypothesis Testing in R

Population mean with Unknown Variance – One Tailed (Traditional Method)

WHAT WE KNOW?

- Sample size, n: 15
- List of data points given
- $H_0: \mu = 0.82$ $H_1: \mu > 0.82$
- Significance level: 0.05

$$T_0 = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

```

1      x = c(0.8411, 0.8191,
2.8182,          0.8125,
3.8750, 0.8580,
4.8532, 0.8483, 0.8276,
5          0.7983, 0.8042,
6.8730,
7          0.8282, 0.8359,
8.8660)
9      n = 15
10     s = sd(x)
11     xbar = mean(x)
12     mu = 0.82
13
14     // Calculate t statistics
15     t = (xbar-mu)/(s/sqrt(n))
16
17     // Calculate critical value
18     alpha = 0.05
19
20     > t
21     [1] 2.718979
22     > t.alpha
23     [1] 1.76131
  
```

Hypothesis Testing in R

Population Variance – (Traditional)

WHAT WE KNOW

- Sample size $n = 15$
- List of data
- $H_0: \mu = 0.82$
- Significance level $\alpha = 0.05$

$T_0 =$

Solution

- Reject H_0 if $t_0 > t_{0.05,14} = 1.761$.
- Computations: Since $\bar{x} = 0.83725$, $s = 0.02456$, $\mu_0 = 0.82$ and $n = 15$, we have

$$t_0 = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{0.83725 - 0.82}{0.02456 / \sqrt{15}} = 2.72.$$

- Conclusion: Since $t_0 = 2.72 > 1.761$, we reject $H_0: \mu = 0.82$ at the 0.05 level of significance that the mean coefficient of restitution exceeds 0.82.

Hypothesis Testing in R

Population mean with Unknown Variance – One Tailed (P-value Method)

WHAT WE KNOW?

- Sample size, n: 15
- List of data points given
- $H_0: \mu = 0.82$ $H_1: \mu > 0.82$
- Significance level: 0.05

pt() is used to get the p-value of T statistics.

2-3 parameters: 1) t statistics, 2) df 3) lower.tail settings

Ref: <http://stat.ethz.ch/R-manual/R-devel/library/stats/html/TDist.html>

```

1      x = c(0.8411, 0.8191,
2            0.8182,          0.8125,
3            0.8750, 0.8580,
4            0.8532, 0.8483, 0.8276,
5            0.7983, 0.8042,
6            0.8730,
7            0.8282, 0.8359,
8            0.8660)
9      n = 15
10     s = sd(x)
11     xbar = mean(x)
12     mu = 0.82
13
14     // Calculate t statistics
15     t = (xbar-mu) / (s/sqrt(n))
16
17     // Calculate p-value of t
18     pval = pt(t, df=n-1,
19               [1] 0.008313369
  
```


Hypothesis Testing in R

Population proportion – One Tailed (Traditional Method)

PROBLEM

Suppose 60% of citizens voted in last election. 85 out of 148 people in a telephone survey said that they voted in current election. At 0.5 significance level, can we reject the null hypothesis that the proportion of voters in the population is below 60% this year?

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

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15
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18
19
20
21
22

Hypothesis Testing in R

Population proportion – One Tailed (Traditional Method)

WHAT WE KNOW?

- Sample size, n: 148
- People who vote : 85
- $H_0: p = 0.6$ $H_1: p < 0.6$
- Significance level: 0.05

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

```
1      n = 148
2      k = 85
3      p = 0.6
4      pbar = k/n
5
6      // Calculate Z statistics
7      z = (pbar-p)/sqrt(p*(1-p)/n)
8
9      // Calculate critical value
10     alpha = 0.05
11     z.alpha = qnorm(1-alpha)
12     -z.alpha
```

```
15
16     > z
17     [1] -0.6375983
18
19     > -z.alpha
20     [1] -1.644854
21
22
```


Hypothesis Testing in R

Population proportion – One Tailed (P-value Method)

WHAT WE KNOW?

- Sample size, n: 148
- People who vote : 85
- $H_0: p = 0.6$ $H_1: p < 0.6$
- Significance level: 0.05

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

```
1      n = 148
2      k = 85
3      p = 0.6
4      pbar = k/n
5
6      // Calculate Z statistics
7      z = (pbar-p)/sqrt(p*(1-p)/n)
8
9      // Calculate p-value for Z
10     pval = pnorm(z)
11
12
13
14
15
16
17
18
19
20
21     [1] 0.2618676
22
```

Hypothesis Testing in R

Population proportion – Two Tailed

PROBLEM

Suppose a coin toss turns up 12 heads out of 20 trials. At .05 significance level, can one reject the null hypothesis that the coin toss is fair?

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22

Hypothesis Testing in R

Population variance – Two Tailed

WHAT WE KNOW?

- Sample size, n : 9
- Sample standard deviation: 3
- $H_0: \sigma = 3.7$ $H_1: \sigma < 3.7$

qchisq() is used to get calculate chi-square statistics

2 parameters: 1) significance level, 2) df

Ref: <http://personal.psu.edu/drh20/astrostatistics/excerpts/html/stats/html/Chisquare.html>

```

1      n = 9
2      s.sd = 3
3      sigma = 3.7
4      alpha = 0.05
5
6      // Calculate chi-square
7      x2 = ((n-
8      ) * s.sd^2) / (sigma^2)
9
10     // Calculate critical value
11     x2.alpha = qchisq(alpha,
12                        n-1)

```

```

> x2
[1] 5.259313
> x2.alpha
[1] 2.732637

```

Hypothesis Testing in R

Population variance – Two Tailed

WHAT WE KNOW?

- Sample size, n: 9
- Sample standard deviation: 3
- $H_0: \sigma = 3.7$ $H_1: \sigma < 3.7$
- Significance level: 0.05

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

```

1      n = 9
2      s.sd = 3
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6      // Calculate chi-square
7      x2 = ((n-
8      ) * s.sd^2) / (sigma^2)
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10     // Calculate critical value
11     x2.alpha = qchisq(alpha,
12     df=n-1)
13
14
15
16     > x2
17     [1] 5.259313
18
19     > x2.alpha
20     [1] 2.732637
21
22
  
```

References

- <http://www.r-tutor.com/>

THE END...